Jingxiang Ma

Contact Information

Department of Mathematics
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Employment

The Ohio State University
Arnold Ross Visiting Assistant Professor — Sept. 2026 – Present

Education

University of Sheffield
Ph.D in Mathematics — Oct. 2022 – July 2026
Supervisor: Andrea Brini
Thesis: *Mirror symmetry and quantum McKay correspondence*

Peking University
M.Sc in Pure Mathematics — Sept. 2019 – July 2022
Supervisor: Huijun Fan
Thesis(in Chinese): *Some studies in Geometric Invariant Theory*

Peking University
B.Sc in Mathematics and Applied Mathematics — Sept. 2015 – July 2019

Research Interests

Broad areas: Algebraic Geometry; Mathematical Physics

Specific interests: Enumerative Geometry; Birational Geometry; Mirror Symmetry

Focused topics: Gromov-Witten Theory; Quantum Cohomology and Frobenius Manifolds; Crepant Resoluition/Transformation Conjecture

Publications and Preprints

Integral and higher-genus quantum McKay correspondence in type D, with Andrea Brini
arXiv:2610.22359 (2026)

On orbifold quantum cohomology of foldings of ADE resolutions
arXiv:2607.11766 (2026)

Involution-equivariant topological recursion and mirror symmetry for the affine binary dihedral Calabi–Yau threefold, with Bohan Fang and Zhuoming Lan
arXiv:2607.07355 (2026)

Dubrovin duality and mirror symmetry for ADE resolutions, with Andrea Brini and Ian Strachan
Proc. Roy. Soc. A. **481**: 20250047 (2025); arXiv:2501.05753

Other writings

Notes for two seminar talks on Quantum Cohomology and Frobenius Manifolds (spring 2025, Sheffield):
On semisimple Frobenius manifolds
Crepant Resolution Conjecture: toric examples

	Notes for two seminar talks on Singularity Theory (autumn 2023, Sheffield): *Multiplication and versal deformation* *Introduction to Saito Frobenius manifolds*	
Research Talks /Presentations	*Mirror symmetry of ADE resolutions, with applications*, Mathematics Seminar Series, Great Bay University	1 Sept. 2026
	On orbifold quantum cohomology of foldings of ADE resolutions, Mathematical Physics Seminar, Wuhan University	12 June 2026
	Introduction to Quantum Cohomology, Junior Geometry Seminar, University of Cambridge	10 Oct. 2025
	Crepant resolutions and curve counting for type D du Val singularities, Algebraic Geometry Seminar(online), Ohio State University	30 Sept. 2025
	Gong Show: *Mirror symmetry for ADE resolutions*, Enumerative Geometry and Geometric Representation Theory, Pisa	23 June - 4 July 2025
	Mirror symmetry for ADE resolutions, New Trends in Moduli, Integrability and Deformations, University of Padova	9-12 June 2025
	Mirror symmetry for ADE resolutions: a Lie theoretic approach, Group Actions and Symplectic Singularities, University of Lille	2-6 June 2025
	Mirror symmetry for ADE resolutions, Calf Seminar, University of Nottingham	24 Mar. 2025
	Dubrovin duality and mirror symmetry for ADE resolutions, Integrable Systems and Mathematical Physics Seminar, University of Glasgow	25 June 2024
	Poster: *Dubrovin duality and mirror symmetry for ADE resolutions*, Integrable Systems: Geometrical and Analytic Approaches, SISSA Trieste	3-7 June 2024
Conferences attended	*UK Algebraic Geometry winter school* University of Lancaster	16-20 Dec. 2024
	Geometry and Physics of Mirror Symmetry University of Sheffield	2-6 Sept. 2024
	Mirrors in the Midlands 2024 University of Birmingham	6-9 Aug. 2024
	Integrability and Moduli University of Lisbon	8-12 July 2024
	String Math 2024 ICTP Trieste	10-14 June 2024
	Derived Categories, Moduli Spaces, and Counting Invariants Imperial College London	3-7 July, 2023

	Derived Categories, Hodge theory and Singularities University of Sheffield	28-30 June 2023
	Moduli spaces: theory and coding Les Diablerets	27 Feb.-3 Mar. 2023
	COW meets EAGLE workshop University of Birmingham	2-3 Feb. 2023
	Structures in Enumerative Geometry University of Sheffield	23-27 Jan. 2023
Teaching and Supervision	Mathematics and Python Programming(CIV109), Analysis and Algebra(MAS2004&MAS2009) Graduate Teaching Assistant at University of Sheffield	spring 2025
	Core Foundation Mathematics(MAS003), Mathematics and Python Programming(CIV109) Graduate Teaching Assistant at University of Sheffield	autumn 2024
	Introduction to Linear Representation Theory of Finite Groups, UGRI summer project supervision at University of Sheffield, with Yoghraj Singh	June and July 2023
Programming	Mathematica (primary); Python; SageMath	
Additional Details	**Citizenship:** Chinese **Languages:** Mandarin Chinese (native), English (fluent)	

Last updated: October 26, 2026